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May 2022 - Present

Papa, Inc. - Staff Software Engineer - Platform team

Technology

- Cloud computing
 - AWS - EC2, S3, RDS, IAM, EKS, ECR, WAF, Lambda, ElastiCache, DocumentDB
- Application Development
 - Ruby, Rails, RSpec, ExUnit, Elixir, Phoenix, PostgreSQL, Redis, MongoDB, Git
- Infrastructure
 - Terraform infrastructure as code on AWS
- Deployment Automation, Scaling, Containerization
 - Kubernetes, Helm, Docker
- CI/CD
 - GitHub Actions

Responsibilities

- Application development
 - Backend application development in Ruby and Elixir environments using Ruby on Rails and Phoenix frameworks in a distributed cloud environment.
- Base layer infrastructure
 - Manage core application clusters, dev/test, staging, production in AWS EKS managed Kubernetes service.
 - Manage datastores: RDS/Postgres, ElastiCache/Redis, DocumentDB/Mongo, S3.
 - Access management: IAM, Okta.
 - Cost management and cost reduction in AWS cloud environment.
 - Manage engineering org's infrastructure via Terraform.
- CI/CD
 - Build and manage CI/CD pipelines using GitHub actions to run automated tests, static analysis, code linting and automated deployments.
- Core languages and libraries
 - Build and manage docker images for the engineering team's core languages including Ruby, Elixir and Erlang.
 - Build application features in Ruby.
- Observability and monitoring
 - Application health visibility.
 - Targeted alerting with Slack integration.
 - Metrics dashboards.

- CI/CD insights: build times, test suite anomalies and regressions.
- Docker images
 - Build production Docker images with multi-stage builds using secure practices.
 - Build Docker images for local development and test.
 - Create and manage AWS ECR repositories.
- Local development environments
 - Build development environments for various engineering teams including application images, databases and storage, workers and communication using Docker Compose.
- Engineering-wide Services
 - Build and maintain the engineering organization's primary deployment and dev-tools service, affectionately known as "Deploybot". This is a Rails web application with integrations to Slack, GitHub, PagerDuty and Shortcut used across all engineering teams.
 - Build and maintain primary Elixir/Phoenix backend application.
 - Functions as deployment management, devex and DORA for production and staging environments.
 - Features include on-demand deployments of an isolated application instance for a specific branch with an anonymized database, automatic redeployment on new commits and Slack based management of config, creation and deletion.
 - Rollback, locking and pausing of deployments using Slack commands.
 - Slack release announcements.
 - Secrets and config management.
 - Tools including PR reports, flakey test notifiers and DORA metrics.

Accomplishments and major projects

- WAF/ALB project
 - Created AWS Application Load Balancers (ALB) with AWS Web Application Firewalls (WAF) for a dozen production services hosted in an EKS cluster.
 - I used Terraform to create the WAFs with custom firewall rules using Kubernetes annotations to associate the WAFs to the ALBs and Helm to deploy the ingress resources.
 - Secured our services from threats including SQL injection, traffic from suspicious IPs and bots.
- Observability and monitoring audit and overhaul
 - Overhauled engineering wide application visibility and monitoring using DataDog.
 - Audited and created alerts for vulnerable areas of the cloud platform including deployment failures, resource usage and infrastructure events.
 - Created dashboards and backing data for flakey tests and regressions that allowed engineering to incrementally improve the speed and reliability of our main automated test suite.
- Feature flagging and A/B testing service

- Built and deployed self-hosted instances of the Growthbook platform used across engineering for feature flags, experiments and A/B testing at the request of the CTO.
- Built a deployment pipeline for isolated, ad hoc Growthbook instances and part of staging deployments (review apps).
- Built a feature flag migration system so that all flags are automatically created in production and staging environments using Elixir that allowed all changes to be tracked in version control, minimizing incidents stemming from feature flag ambiguity.
- Database visibility initiative
 - Built pgweb PostgreSQL client into all production and staging deployments and local development environments for database visibility.
 - Built secure Docker image, created Helm charts and Kubernetes ingress resources and associated load balancers.
 - Created read-only roles and policies allowing product and engineering to safely interact with production databases for reporting and insights.
- Deploybot proxy for platform development
 - Created a workflow in the Platform team's central Rails application that allows the team to develop and test the most important features of the application in a production-like way which includes auth backed by GitHub SSO, GitHub event ingestion, Slack app integration and various command handlers.
 - This allowed the Platform team to simultaneously develop and test core features with minimal setup effort.

May 2020 - February 2022

Red Canary, Denver, CO - Staff Software Engineer - Technical Lead

Technology

- Cloud Computing
 - AWS, EC2, s3, IAM, STS, Cognito, API Gateway, SQS, SNS, Lambda, RDS, DynamoDB, ECR container registry, some Azure
- Application Development
 - Ruby, JRuby, Ruby on Rails, Postgres/SQL, Redis, Apache Kafka event streams, Hotwire, React, HTML, Protocol Buffers/Protobuf, Git/Github, Elixir
- Deployment/DevOps
 - Docker, Kubernetes, Terraform, Helm, Circle CI, Argo CD, event driven scaling
- Monitoring/Metrics
 - Prometheus, Grafana, CloudWatch, Sentry

Responsibilities/Accomplishments

- Full stack feature development in various Rails apps and stand alone Ruby components at scale in a distributed cloud computing environment with excellent test coverage.
- Built CI/CD pipelines running automated test suites, Docker builds and deploys.
- Lead architectural design of new products.
- Develop RESTful microservices.
- Build computing infrastructure using Terraform to support cloud based applications.
- Provide technical leadership for various engineering teams.
- Plan and develop backend data integrations with cloud native SIEM, EDR, network security and alerting products using Ruby/Rails solutions in a distributed cloud computing environment.
- Ensure the security of tens of thousands of customer endpoints from cyber threats, ransomware, mitre attacks, etc.
- Develop event driven architecture data pipelines using Apache Kafka.
- Lead sprint planning and backlog refinement in an Agile environment.
- Mentorship and development of junior engineers through pair programming, extensive PR review and TDD.
- Host an engineering book club.
- Provide continued support and maintenance to our ingestion and data pipeline systems including regular PagerDuty on-call shifts, incident management and remediation and incident retrospectives.
- Write detailed documentation of engineering systems and procedures.
- Lead 1-2 technical interviews per week, helping double the engineering organizations' capacity over the last 2 years.
- Build monitoring and alerting dashboards with Prometheus metrics and Grafana that allow deep insights into a complicated distributed system with minimal downtime using a data driven approach.
- Regularly participate in our internal architectural design committee to pitch and plan large scale, system wide changes to support new data processing pipelines and new functionality in the system at large involving technical writing, public speaking, presenting solutions, etc.
- Cross team projects and integration in a complex environment:
 - 400-1000 EC2 instances running at any given time
 - 50k total cores, 100+ TB of memory, running 5-10k pods
 - Ingesting 1PB of data into s3 daily, growing by 1TB/day
 - Running 16 Postgres Databases in RDS
 - Running 4 Kubernetes clusters in multiple VPCs

TextUs, Boulder, CO - Sr. Software Engineer

Technology

- Ruby, Ruby on Rails, RSpec, React, PostgreSQL, AWS, SQS, Heroku, GitHub, Elasticsearch, CircleCI, Javascript, React

Responsibilities/Accomplishments

- Develop full stack features in a customer facing Rails Application.
 - Architect primary application components.
 - Built an integration for a strategic partner company including OAUTH registration, token persistence, refresh and application logic.
 - Built an ingress message handling system to accept delivery receipts and inbound SMS and MMS text message webhooks from various service providers.
 - Build an egress messaging system to dispatch text messages, communicating with our SMS provider's REST API's.
 - Built a "V2" application from the ground up while migrating customers from the legacy app.
 - Built an analytics engine to support account and user statistical data and exporting.
 - Object Oriented Design, automated testing and TDD
 - Pair programming with and mentorship of junior developers.
 - Managing epic and user story creation, planning, and execution.
 - Application monitoring with Papertrail, Honeybadger, etc.
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Aug 2017 - May 2018

Kapost, Boulder, CO - Sr. Software Engineer

Technology

- Ruby, Rails, RSpec, React, Backbone, MongoDB, PostgreSQL, Elasticsearch, GitHub, CircleCI

Accomplishments and Responsibilities

- Full stack development for a customer facing Rails application.
- Improved the health of our test suite and increased coverage for some mission critical areas of the application.
- Dramatically increased the consistency of our CI platform (CircleCI) and reduced false negative test failures.
- Mentorship and pair programming with junior developers on basic principles of Object Oriented Design and testing.
- Built a content tagging feature for a primary customer and delivered ahead of schedule.

- Reduced memory consumption and increased request throughput on our Heroku dynos by using jemalloc for memory allocation and fine tuning our Ruby GC constants.

Aug 2014 - Aug 2017

Goldstar Learning, Chicago, IL - Software Engineer

Technology

- Ruby, Rails, RSpec, EmberJS, PostgreSQL, AWS, ECR, Docker, Docker compose, CircleCI, Perl

Responsibilities/Accomplishments

- Develop full-stack features for a customer facing Rails application.
- Broke apart and rewrote an untested, tightly coupled, legacy Perl web application into a loosely coupled, dockerized Rails application with over 95% test coverage using RSpec.
- Designed a new REST api to serve a heavily used "item-bank", from redesigning the database layer (PostgreSQL), to writing the application code (Ruby/Rails/EmberJS).
- Wrote a feature rich front end application in EmberJS for our new "item-bank" product feature.
- Wrote a new authentication and authorization service which allowed us to integrate our legacy application with new web services and external tools, issue and manage JSON Web Tokens (JWTs) and to route traffic to customers in different environments, e.g. AWS and a separate collocated datacenter, whilst maintaining 95% test coverage.
- Mentored and pair-programmed with junior engineers on fundamentals of test driven development and object oriented design, dependency injection, duck-typing and loose coupling.

2013

Various contract web development

2012

United States Navy - Honorably Discharged

Education

University of Illinois, Urbana-Champaign - Bachelor's degree